# EESC3220 (Spring 2026)
# Atmospheric Chemistry

**Atmospheric chemistry** is a study of how a molecule introduced into the atmosphere is altered by the oxidizing medium of the atmosphere and, in turn, how this alteration affects the atmospheric composition and atmospheric properties.

## 1 Introduction

### Air pollution

- The introduction of chemicals, particulate matter, or biological materials that cause harm or discomfort to humans or other living organisms, or cause damage to the natural environment or built environment, into the atmosphere.
- HKEPD has established air quality standards for six of the most common air pollutants - carbon monoxide, lead, ozone, particulate matter, nitrogen dioxide, and sulfur dioxide - known as "criteria" air pollutants. HKEPD sets secondary Air Quality Objectives (AQO) to protect public welfare from adverse effects of criteria pollutants, including protection against visibility impairment, or damage to animals, crops, vegetation, or buildings.
- Air Quality Health Index (AQHI) is a health risk-based air pollution index. It provides an estimate of the short-term (hospital admission) risk of heart and respiratory diseases from air pollution.

### Ozone

- Ozone is a colorless gas.
- Good Ozone is present naturally in the Earth's upper atmosphere—10 to 30 miles above the Earth's surface. This natural ozone shields us from the sun's harmful ultraviolet rays. Peak O3 is about 12ppm at about 20km.
- Bad Ozone forms near ground level when air pollutants (emitted by sources such as cars, power plants, and chemical plants) react chemically in the presence of sunlight. Created by photochemical reactions between NOx and CO, VOCs in the presence of heat and sunlight.
- Ozone can irritate our respiratory system and reduce lung function.

### Particulate Matters (PM)

- Particulate matter is a mixture of solid particles and liquid droplets in the air, including dust, soot, and smoke, which poses significant health risks. Ranging from coarse ($\mathrm{PM}_{10}$) to fine ($\mathrm{PM}_{2.5}$) particles, these pollutants originate from human activities like combustion and natural sources like wildfires.
- The number followed by PM refers to the particle with diameter smaller than the number in micrometers.
- $\mathrm{PM}_{10}$ are usually found near roads and industries. $\mathrm{PM}_{2.5}$ is especially dangerous, as it can penetrate deep into the lungs and bloodstream, causing cardiovascular and respiratory diseases.

### Acid Rain

- Rain water with $\mathrm{pH} < 5$.

## 2 Measures of Atmospheric Composition

### Composition of Atmosphere (at 1 atm, 288 K)

| Gas | Mixing Ratio (mol/mol) | Gas | Mixing Ratio (mol/mol) |
|---|---|---|---|
| $N_2$ | $0.78$ | Formaldehyde | $3.3 \times 10^{-9}$ |
| $O_2$ | $0.21$ | Acetaldehyde | $1.1 \times 10^{-9}$ |
| Ar | $0.0093$ | Benzene | $550 \times 10^{-12}$ |
| $CO_2$ | $365 \times 10^{-6}$ | Isoprene | $500 \times 10^{-12}$ |
| CO | $(0.1\text{-}20) \times 10^{-6}$ | $\alpha$-pinene | $300 \times 10^{-12}$ |
| $O_3$ | $(0.01\text{-}10) \times 10^{-6}$ | CFC-12 | $350 \times 10^{-12}$ |
| $CH_4$ | $1.72 \times 10^{-6}$ | CFC-11 | $200 \times 10^{-12}$ |
| $N_2O$ | $310 \times 10^{-9}$ | | |

### VOC (Volatile Organic Compounds)

- High-vapor-pressure, carbon-based chemicals (e.g., methane, toluene, isoprene) emitted into the atmosphere from natural (vegetation, wildfires) and anthropogenic (fuels, solvents) sources.

## Gas Phase Oxidants: Radicals

- Radicals are species with an unpaired e$^-$.
- It is energized with high tendency to pair the electron to reduce its free energy.
- Reactions are often rapid.
- The hydroxyl radical ($\cdot$ OH) is the primary daytime oxidant in the troposphere, whereas the nitrate radical ($\cdot$ $NO_3$) dominates during nighttime by the reaction of $NO_2$ and $O_3$.

## Mean Molecular Weight of Dry Air

$$M_a = \sum_i C_i M_i$$

## Mixing Ratio

$$C_X = \frac{\text{\# moles of X}}{\text{\# moles of air}}$$

- In units of mol/mol, equivalently in units of v/v. Usually used for gaseous constituents *only*.
- ppm (parts per million= $10^{-6}$ ppm), ppb (parts per billion= $10^{-9}$ ppm), ppt (parts per trillion= $10^{-12}$ ppm)
- Mixing ratio of a gas remains constant when density changes.

## Number Density

$$n_X = \frac{\text{\# of moledules}}{\text{volume of air}}$$

- In units of $\text{molecules cm}^{-3}$. Can be used for gaseous constituents and aerosols.
- The proper measure for calculating reaction rates (chemical kinetics) and optical properties of the atmosphere.
- There are ~$2.55 \times 10^{19}$ molecules in one $\text{cm}^3$ of air under room temperature and pressure conditions.
- Atmospheric column determines the total efficiency with which the gas absorbs or scatters light passing through the atmosphere.

$$\Omega_X = \int_0^\infty n_X(z)\,\mathrm{d}z$$

- The number density and the mixing ratio of a gas are related by the number density of air $n_a$.

$$n_X = C_X n_a = C_X \frac{A_v P}{RT}$$

## Mass Concentration

$$\rho_X = \frac{\text{mass of species}}{\text{volume of air}} = \frac{n_X M_X}{A_v}$$

- In units of $\mu\text{g/m}^3$ or $\text{ng/m}^3$. Can be used to measure particulate matters.

## Dobson Unit and Ozone Column

- One Dobson unit represents a 0.01 mm thick layer of ozone under STP.
- Total column ozone refers to the total amount of ozone in a vertical column of the atmosphere from the Earth's surface to the top of the atmosphere.
- Healthy ozone profile is approximately 3DU of ozone, representing a total ozone layer thickness of roughly 3mm if compressed to sea-level pressure and temperature.
- $1\ \text{DU} = 2.69 \times 10^{16}\ \text{molecules cm}^{-2}$.

## Vapor Pressure

- Raoult's Law: $P_X = C_X P_X^\circ$. Vapor pressure of solution is equal to pressure of pure solvent multiplied by its mole fraction. Hence, the presence of CCNs make clouds easier to form.
- Clausius-Clapeyron equation: $e_s = 6.108 e^{0.0668T}$.
- Relative Humidity (RH)

$$\text{RH} = \frac{e}{e_s} \times 100\%$$

- Specific Humidity

$$q_s \approx 0.622 \frac{e}{p}$$

Conserved as the air parcel moves, as long as water vapor does not condense.

### Exercise 2.1

Estimate the mass of $\mathrm{CO_2}$ in the atmosphere, assuming that the mass of the atmosphere is $m_a = 5.2 \times 10^{18}$ kg and the mixing ratio of $\mathrm{CO_2}$ is $400$ ppm.

*Solution*. $m_{\mathrm{CO_2}} = \frac{m_a}{M_a} \cdot C_{\mathrm{CO_2}} \cdot M_{\mathrm{CO_2}} = \frac{5.2\times10^{18}\ \mathrm{kg}}{0.029\ \mathrm{kg\ mol^{-1}}} \times 400 \times 10^{-6}\ \mathrm{mol\ mol^{-1}} \times 0.048\ \mathrm{kg\ mol^{-1}} = 3.5 \times 10^{15}\ \mathrm{kg}.$

## 3 Atmospheric Structure and Chemical Thermodynamics

### Troposphere

- Extending from surface to about 7-15 km high, ending at the tropopause boundary.
- Relatively linear decrease in temperature with height up to the troposphere (with lapse rate $\Gamma = \mathrm{d}T/\mathrm{d}z = 6.5\ \mathrm{K/km}$).

### Stratosphere

- Extends to about 50 km
- Temperature inversion: rises with increasing altitude to a maximum of about $-1\ ^\circ\mathrm{C}$ at the stratopause, stable.

### Measuring Atmospheric Pressure

- Weight exerted by the overhead atmosphere on a unit area of surface.
- A long glass tube filled with mercury inverted over the pool creates a vacuum in head space with equilibrium height $h$ obtains the atmospheric pressure $P_A$ by

$$P_A = \rho_{\mathrm{Hg}} g h,$$

  where $\rho_{\mathrm{Hg}} = 13.6\ \mathrm{g\ cm^{-3}}$ is the density of mercury and $g = 9.8\ \mathrm{m\ s^{-2}}$ is the acceleration of gravity.
- $h = 76\ \mathrm{mmHg}$, which converts to $P_A \approx 101300\ \mathrm{Pa}$.
- Take $R = 6400\ \mathrm{km}$, then the mass of the atmosphere is given by

$$m_a = \frac{4\pi R^2 P_S}{g} = 5.2 \times 10^{18}\ \mathrm{kg}$$

### Barometric Law

$$P(z) = P(0)e^{-z/H}, n_a(z) = n_a(0)e^{-z/H}, \text{where } H = \frac{RT}{M_a g}$$

- **Scale height**: a characteristic vertical distance where the pressure drops by a factor of $e$, representing how fast the pressure is decaying as height increases.

### Land-Sea Breeze Effect

- Air sea-land interaction induced by differential heating on land and sea surface.
- **Mechanism**: Land heats more than sea$\Rightarrow T_L > T_S \Rightarrow H_L > H_S \Rightarrow P_L(z) > P_S(z) \Rightarrow$high-altitude flow from land to sea$\Rightarrow$reverse surface flow from sea to land.

### Physical Chemistry

- Bond formation involves valence electrons (outer shell).
- Octet Rule: Atoms (other than H and He) tend to bond to achieve 8 valence electrons. Hydrogen seeks 2.
- Bond breaking requires energy input, bond formation releases energy.
- Ionic Bonds: Formed by the electrostatic attraction between positive and negative ions. This involves a complete transfer of electrons (e.g., $\mathrm{Na^+}$ and $\mathrm{Cl^-}$).
- Polar Covalent Bonds: Formed when two different atoms share electrons unequally due to a difference in electronegativity. This creates slight positive and negative charges on the ends of the bond.
- **Radicals**: species with an unpaired electron. High energy and highly reactive due to the strong tendency to pair the unpaired electron. They drive the vast majority of gas-phase chemical reactions in the atmosphere.
- **Oxidation**: Loss of electrons. The oxidation state (O.N.) increases.
- **Reduction**: Gain of electrons. The oxidation state decreases.
- **Oxidation State** (O.N.): A measure of the degree of oxidation of an atom. A higher positive number means more oxidized; a higher negative number means more reduced.
- Enthalpy: thermodynamic potential of the system. $\Delta H > 0$: endothermic; $\Delta H < 0$: exothermic.
- $\Delta H_f^0$: standard heat of formation (298 K, 1 atm).
- Hess Law: $\Delta H_{r\times n}^0 = \sum \Delta H_f^0\ (\text{products}) - \sum \Delta H_f^0\ (\text{reactants})$
- **Entropy**: a measure of randomness or disorder of a system. System tends to evolve toward a state of greater disorder.

$$\Delta S_{r\times n}^0 = \sum \Delta S^0\ (\text{products}) - \sum \Delta S^0\ (\text{reactants})$$

- **Gibbs Free Energy**: $\Delta G_{r\times n}^0 = \Delta H_{r\times n}^0 - T\Delta S_{r\times n}^0$. If $\Delta G < 0$, the reaction is spontaneous; If $\Delta G = 0$, the system is at equilibrium; If $\Delta G < 0$, the reaction is non-spontaneous.
- **Thermodynamics** ($\Delta$G) describes if a reaction can happen. **Kinetics** describes how fast it will happen.

# 4 Chemical Kinetics and Atmospheric Photochemistry

## Atmospheric Reactions

- **Thermal**: collision of molecules or internal vibrations, including decomposition, combination, and disproportionation.
- **Photochemical**: absorption of a photon provides energy for reaction, including collisional deactivation, photodissociation, intramolecular rearrangement, photoisomerization, and direct reaction.

## Order of Reactions

- **First-order reaction (unimolecular)**: $A \to B + C$

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = -\frac{\mathrm{d}[B]}{\mathrm{d}t} = -\frac{\mathrm{d}[C]}{\mathrm{d}t} = -k_1[A]$$

  The first-order rate coefficient $k_1$ has unit of $\mathrm{s}^{-1}$.
- **Second-order reaction (bimolecular)**: $A + B \to C + D$

$$\frac{\mathrm{d}[A]}{\mathrm{d}t} = \frac{\mathrm{d}[B]}{\mathrm{d}t} = -\frac{\mathrm{d}[C]}{\mathrm{d}t} = -\frac{\mathrm{d}[D]}{\mathrm{d}t} = -k_2[A][B]$$

  The second-order rate coefficient $k_2$ has unit of $\mathrm{cm}^3\ \mathrm{molecule}^{-1}\ \mathrm{s}^{-1}$.
- **Third-order reaction (termolecular)**: $A + B + M \to AB + M$
  First, molecules $A$ and $B$ collide to produce an energetic intermediate $AB^*$. Then, the excess energy is removed by collision with another molecule denoted by $M$. The third body $M$ is any inert molecule that dissipate the excess energy as heat (e.g. $\mathrm{N}_2, \mathrm{O}_2$ in the atmosphere), stabilizing the excited product.
  Mechanism:
  $(\mathrm{R1})\ A + B \to AB^*$
  $(\mathrm{R2})\ AB^* \to A + B$
  $(\mathrm{R3})\ AB^* + M \to AB + M^*$
  $(\mathrm{R4})\ M^* \to M + \mathrm{heat}$

$$k_1[A][B] = k_2[AB^*] + k_3[AB^*][M] \Rightarrow [AB^*] = \frac{k_1[A][B]}{k_2 + k_3[M]}$$

$$\frac{\mathrm{d}[AB]}{\mathrm{d}t} = k_3[AB^*][M] \Rightarrow \frac{\mathrm{d}[AB]}{\mathrm{d}t} = \frac{k_1 k_3[A][B][M]}{k_2 + k_3[M]}$$

## Photolysis

$$J_X = q_X \sigma_X I, k = \int_{\lambda_0}^{\lambda_1} q_X(\lambda)\sigma_X(\lambda) I_\lambda\ \mathrm{d}\lambda$$

$q_X$: quantum yield $(\mathrm{molecules\ photon}^{-1})$, ranging from 0 to 1, probability of photolysis of $X$;
$\sigma_X$: absorption cross-section in units of $\mathrm{cm}^2\ \mathrm{molecule}^{-1}$;
$I$: number of photons crossing a unit horizontal area per unit time from any direction $\mathrm{photons\ cm}^{-2}\ \mathrm{s}^{-1}$.

## Bimolecular Thermal Reactions Rates

- **Arrhenius Equation**: $k = Ae^{-E_a/RT}$
- **A-factor**: molecular cross-section x mean relative collision velocity x steric factor

$$A_{\max} = \pi(r_A + r_B)^2 \left(\frac{8k_B T}{\pi(m_A + m_B)}\right)^{1/2}$$

  Typically, $d\sim 3 \times 10^{-8}\ \mathrm{cm}, v\sim 100\ \mathrm{m/s}$, hence $A_{\max} \approx 2.5 \times 10^{-10}\ \mathrm{cm}^3\ \mathrm{molecule}^{-1}\ \mathrm{s}^{-1}$.
  The steric factor describes the probability of having the reactive parts of the molecules facing each other during collision.

## Three Body Reactions: High-Pressure and Low-Pressure Limit

- The formation of $AB$ is pressure-dependent (for the concentration of $M$).
- Low pressure limit: $[M] \ll k_2/k_3$, then

$$\frac{\mathrm{d}[AB]}{\mathrm{d}t} = \frac{k_1 k_3[A][B][M]}{k_2} = k_o[A][B][M]$$

- High pressure limit: $[M] \gg k_2/k_3$, then

$$\frac{\mathrm{d}[AB]}{\mathrm{d}t} = k_3[A][B]$$

- General case: let $k_\infty := k_3$, $k_o := k_1 k_3/k_2$, then

$$\frac{\mathrm{d}[AB]}{\mathrm{d}t} = \frac{k_o[A][B][M]}{1 + (k_o/k_\infty)[M]}$$

### Major Photolysis Examples

- $O_3$ photolysis:
  $\lambda < 320$ nm: $O_3 + h\nu \to O(^1D) + O_2$ (produces excited oxygen, a major source of OH).
  $\lambda \geq 320$ nm: $O_3 + h\nu \to O(^3P) + O_2$
- $NO_2$ photolysis ($\lambda < 397.8$ nm): $NO_2 + h\nu \to NO + O(^3P)$.
  *Critical step*: primary source of ozone in the troposphere (the $O(^3P)$ atom quickly combines with $O_2$ to form $O_3$).
- HONO photolysis: $HONO + h\nu \to OH + NO$ (major morning source of OH).
- HCHO photolysis: $HCHO + h\nu \to H + HCO$ or $H_2 + CO$.
- $H_2O_2$ photolysis: $H_2O_2 + h\nu \to 2OH$.

### Chemical Family

- Two species interconvert rapidly (reversibly) compared to other loss processes.

### Pseudo-Steady State Approximation

- For highly reactive intermediates (like radicals) that have very short lifetimes.
- **Assumption**: The intermediate is produced and consumed so quickly that its concentration does not change over time ($\mathrm{d}[\text{intermediate}]/\mathrm{d}t = 0$), rate of generation of the intermediate is equal to the rate of disappearance.

## 5 Solar Radiation Absorption by Gas Molecules

### Atmospheric Radiation and Absorption

- The solar spectrum at the top of the atmosphere is filtered by the absorption by $O_2$, $O_3$ and transformed into the jagged, filtered spectrum. At surface, wavelength > 290nm. $\lambda < 240$ nm: absorption by $O_2$; $\lambda < 320$ nm: absorption by $O_3$.
- **UV/Visible Absorption**: Causes changes in the electronic state of molecules. This is high-energy absorption that can lead to photodissociation (breaking bonds).
- **IR Absorption**: Causes changes in the vibrational and rotational states of molecules. This requires the molecule to have an electric dipole moment. This is the basis of the greenhouse effect (trapping heat).
- **Absorption by $O_2$** – confined to very short wavelengths ($< 250$ nm), dissociative
- **Absorption by $O_3$**
  - Hartley band ($200 - 300$ nm, strongest absorption region $\sigma \sim 10^{-17} - 10^{-18}$ cm$^2$), causes temperature inversion in the stratosphere.
  - Huggins band ($300 - 360$ nm), allows some UV to penetrate to the troposphere.
  - Chappuis Band ($450 - 650$ nm, visible light, $\sigma \sim 10^{-21}$ cm$^2$).
- **Actinic Flux** ($I(\lambda)$): The number of photons available for photochemistry ($\text{photons cm}^{-1}\text{s}^{-1}\text{nm}^{-1}$).

### Beer-Lambert Law and Optical Depth

$$\frac{\mathrm{d}F(x,\lambda)}{\mathrm{d}x} = -b(x,\lambda)F(x,\lambda), b = b_{\text{absorption}} + b_{\text{scattering}}$$

$$\tau(x_1, x_2; \lambda) = \int_{x_1}^{x_2} b(x,\lambda)\ \mathrm{d}x$$

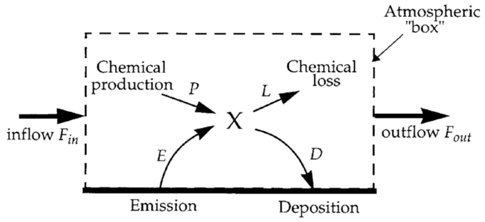
- Solar irradiance: $F(x,\lambda)$, in unit of $\text{photons cm}^{-2}\ \text{s}^{-1}$, Extinction coefficient: $b(x,\lambda)$, in unit of $\text{cm}^{-1}$.
- Optical depth: $\tau(x_1, x_2; \lambda)$, an integral of extinction coefficient up to a specific wavelength.
- At the attitude at which $\tau = 1$, the radiation is reduced to $1/e$ of its value at the top of the atmosphere.
- Molecular absorption $b_a$ of gas $A$: $b_a = n_A \sigma_A(\lambda)$, where $\sigma_A$ is the absorption cross-section of gas $A$ at wavelength $\lambda$.
- If the vertical optical depth is $\tau_{\text{vertical}}$, then at solar zenith angle $\theta_0$, $\tau(\theta_0) = \tau_{\text{vertical}} \cdot \sec\theta_0$, $F = F_0 \exp(-\tau_{\text{vertical}} \sec\theta_0)$.

## 6 Atmospheric Lifetime and Simple Models

### Four Types of Processes:

- **Emissions**: anthropogenic fossil fuel combustion, aircrafts, biogenic photosynthesis of oxygen, nonbiogenic volcanoes
- **Chemistry**: Reactions in the atmosphere can lead to the formation and removal of species.
- **Transport**: Winds transport atmospheric species away from their point of origin.
- **Deposition**: All material in the atmosphere is eventually deposited back to the Earth's surface. Escape from the atmosphere to outer space is negligible because of the Earth's gravitational pull.
  - Dry deposition involves direct reaction or absorption at the Earth's surface, such as the uptake of CO by photosynthesis
  - Wet deposition: involves scavenging by precipitation.

## One-Box Model



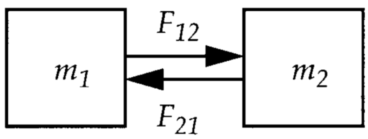

$$\frac{\partial[X]}{\partial t} = E_X - \nabla \cdot (\mathbf{U}[X]) + P_X - L_X - D_X$$

- Assumption: $X$ is well-mixed in the box, no spatial distribution.
- Lifetime: $\tau = m/(F_{\text{out}} + L + D)$.
- For first-order chemical loss $L = k_c[A]$, $\tau_A = [A]/k_c[A] = 1/k_c$;
- For second-order chemical loss $\tau_A = [A]/k[A][B] = 1/k[B]$.
- Fraction removed by export: $f = \frac{F_{\text{out}}}{F_{\text{out}}+L+D}$; Export by $x$-direction wind of speed $U$: $\tau_{\text{out}} = \frac{\rho_X l_x l_y l_z}{\rho_X U l_y l_z}$.
- Assuming first-order sink, then $\frac{\partial m}{\partial t} = S - km$, i.e. sources are independent of $m$, sinks are first-order in $m$. Hence,

$$m(t) = m(0)e^{-kt} + \frac{S}{k}(1 - e^{-kt}).$$

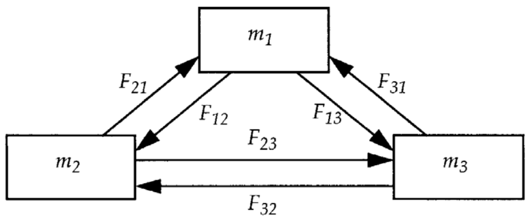
- $\tau$: $e$-folding time to measure the time that it takes for the system to reach steady state, decay of initial condition to $1/e$ $(37\%)$ of the initial state, $1 - 1/e$ $(63\%)$ of the final value.

## Two-Box Model




$$\frac{\partial m_1}{\partial t} = E_1 + P_1 + F_{21} - L_1 - D_1 - F_{12}, \ \frac{\partial m_2}{\partial t} = E_2 + P_2 + F_{12} - L_2 - D_2 - F_{21}$$

## Three-Box Model




Lifetimes: $\tau_1 = \frac{m_1}{L_1+D_1+F_{12}+F_{13}}, \tau_{12} = \frac{m_2}{L_1+L_2+D_1+D_2+F_{13}+F_{23}}$

## Eulerian Model

- Described by a stationary 3D grid (or mesh) in fixed boxes over the entire globe.
- The model then calculates how the concentration of chemicals within each specific box changes over time.
- Best used for urban air quality forecasting, regional haze studies, and global climate modeling (e.g., models, e.g. GEOS-Chem or CMAQ).

## Lagrangian ("Puff") Model

- The model follows the air parcel moving with wind.
- A continuous release of smoke from a stack is treated as a series of discrete puffs released at regular time intervals.
- The transport terms are implicit in the trajectory.
- In pollution plume:

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = E + P - L - D - k_{\text{dilution}}([X] - [X]_b)$$

## Column Model

- Assumes the atmospheric conditions (like wind and concentration) are horizontally uniform across a large area.
- The temperature inversion defines the mixing depth, deeper during daytime due to more turbulence.

- Best used for studying ozone formation in the boundary layer, testing new chemical mechanisms, understanding aerosol-cloud interactions, and benchmarking complex chemistry before putting it into a 3D model.
- In column moving across the city:

$$\frac{\mathrm{d}[X]}{\mathrm{d}t} = \frac{E}{h} - k[X] = \frac{\mathrm{d}[X]}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = U\frac{\mathrm{d}[X]}{\mathrm{d}x}$$

### Exercise 6.1

What is the difference between the e-folding lifetime $\tau$ and the half-life $t$ of a radioactive element?

*Solution.* Consider a radioactive element $X$ decaying with a rate constant $k$ ($s^{-1}$), $\mathrm{d}[X]/\mathrm{d}t = -k[X]$. Then, $[X](t) = [X](0)e^{-kt}$. The half-life is $t_{1/2} = (\ln 2)/k$. The e-folding time is the time at which 37% of $[X](0)$ remains: $\tau = 1/k$. Hence, $t_{1/2} = \tau \ln 2 = 0.69\tau$.